Blessed G/Michael Catholic Secondary School

Semester 1 Chemistry Mid1 Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** What is the general molecular formula of saturated hydrocarbons (Alkanes)?

A. CnH2n **B.** CnH2n+2 **C.** CnH2n-2 **D.** CnH2n-6

____ **2.** What is the hybridization of carbon atoms in an alkane molecule like methane (CH4)?

A. sp **B.** sp2 **C.** sp3 **D.** dsp2

____ **3.** What is the IUPAC name for the hydrocarbon CH3-CH(CH3)-CH2-CH3?

A. 2-methylpropane **B.** 2-methylbutane

C. Isopentane **D.** Dimethylpropane

____ **4.** How many structural isomers does butane (C4H10) have?

A. 1 **B.** 2 **C.** 3 **D.** 4

____ **5.** Complete combustion of any hydrocarbon in excess oxygen produces:

A. CO and H2 **B.** CO2 and H2O **C.** C and H2O **D.** CO2 and H2

____ **6.** Which of the following hydrocarbons contains a triple bond between carbon atoms?

A. Ethene **B.** Ethane **C.** Ethyne **D.** Propene

____ **7.** The reaction between methane and chlorine in the presence of ultraviolet light is an example of:

A. Addition reaction **B.** Substitution reaction

C. Elimination reaction **D.** Polymerization

____ **8.** According to Markovnikov’s rule, when HBr adds to propene (CH3-CH=CH2), the hydrogen atom attaches to:

A. The carbon with fewer hydrogen atoms **B.** The carbon with more hydrogen atoms

C. The central carbon exclusively **D.** Any carbon randomly

____ **9.** What type of isomerism is exhibited by but-2-ene in cis and trans arrangements?

A. Chain isomerism **B.** Positional isomerism

C. Geometric (stereoisomerism) **D.** Functional group isomerism

____ **10.** Which reagent is used as a test for unsaturation (decolorized by alkenes and alkynes)?

A. Bromine water (Br2 in CCl4) **B.** Dilute hydrochloric acid

C. Sodium hydroxide solution **D.** Litmus indicator